

Graduation Project (Group 3):
Building and Operating a Mini Security
Operations Center (SOC)

Prepared by:

طه محمود عبدالرحمن

محمود ابوهيبه راغب

نرمين محمد عبدالمنعم

يوسف حماده يوسف

ابانوب عزيز امين

مصطفى عبدالسميع

Phase 1: SOC Setup & Log Ingestion

During this phase, a Security Operations Center (SOC) lab environment was established using three virtual machines deployed via Oracle VirtualBox. The environment was designed to simulate real-world attack and monitoring scenarios.

Virtual Machine 1 – SIEM Server (Ubuntu Server – IP: 192.168.244.132)

The first virtual machine is based on an Ubuntu Server operating system. This machine serves as the central SIEM server with Splunk Enterprise installed.

It is responsible for:

- Receiving logs from endpoint systems
- Indexing and storing security events
- Enabling real-time monitoring and analysis

This system acts as the core component of the SOC environment.

Virtual Machine 2 – Client Machine (Windows 10 – IP: 192.168.244.134)

The second virtual machine runs a Windows 10 (x64) operating system and represents the victim or client system targeted during the attack simulation.

This machine has Splunk Universal Forwarder installed, which is configured to:

- Collect system and security logs

- Forward logs to the central SIEM server (Ubuntu machine)

This allows visibility into activities occurring on the endpoint.

Virtual Machine 3 – Attacker Machine (Kali Linux – IP: 192.168.244.136)

The third virtual machine is based on Kali Linux and is used as the attacker system.

This machine is utilized to:

- Perform reconnaissance using tools like Nmap
 - Identify vulnerabilities in the target system
 - Launch attacks and deliver payloads
-

Network Configuration

All systems are configured within the same private network range:

192.168.244.130 – 192.168.244.140

This ensures direct communication between:

- The attacker (Kali Linux)
 - The victim (Windows 10)
 - The SIEM server (Ubuntu with Splunk Enterprise)
-

Log Flow Architecture

The log ingestion pipeline is configured as follows:

Windows 10 (Forwarder) → Splunk Enterprise (Ubuntu SIEM Server)

- Logs generated on the client machine are collected by the forwarder
- Logs are sent securely to the SIEM server over port 9997
- The SIEM platform processes and stores the logs for further analysis

Phase 2: SIEM Configuration & Use Case Development

This section describes the implementation of the Security Operations Center (SOC) components, including the installation and configuration of the SIEM platform (Splunk) and the Splunk Universal Forwarder for log collection.

4.4.1 Installing SIEM (Splunk)

The following steps describe the installation of Splunk Enterprise on a Debian-based Linux system:

Step 1: Download Splunk Package

```
wget -O splunk-9.4.1-e3bdab203ac8-linux-amd64.deb \
```

```
"https://download.splunk.com/products/splunk/releases/9.4.1/linux/splunk-9.4.1-e3bdab203ac8-linux-amd64.deb"
```

Step 2: Install Splunk

```
sudo dpkg -i splunk-9.4.1-e3bdab203ac8-linux-amd64.deb
```

Step 3: Start Splunk Service

```
sudo /opt/splunk/bin/splunk start --accept-license --answer-yes --no-prompt  
--seedpasswd 80808080
```

This command starts Splunk for the first time, accepts the license automatically, and sets the administrator password.

Step 4: Enable Auto Start on Boot

```
sudo /opt/splunk/bin/splunk enable boot-start
```

Step 5: Access Splunk Web Interface

Splunk can be accessed via the following URL:

<http://localhost:8000>

4.4.2 Installing Splunk Universal Forwarder (Linux)

The Splunk Universal Forwarder is used to collect logs from endpoints and forward them to the Splunk server.

Step 1: Install Splunk Forwarder (RPM-based systems)

```
sudo rpm -ivh splunkforwarder-9.4.1-e3bdab203ac8.x86_64.rpm
```

Step 2: Start the Forwarder

```
sudo /opt/splunkforwarder/bin/splunk start --accept-license
```

During the first startup, you will be prompted to set the administrator username and password.

Step 3: Enable Auto Start on Boot

```
sudo /opt/splunkforwarder/bin/splunk enable boot-start
```

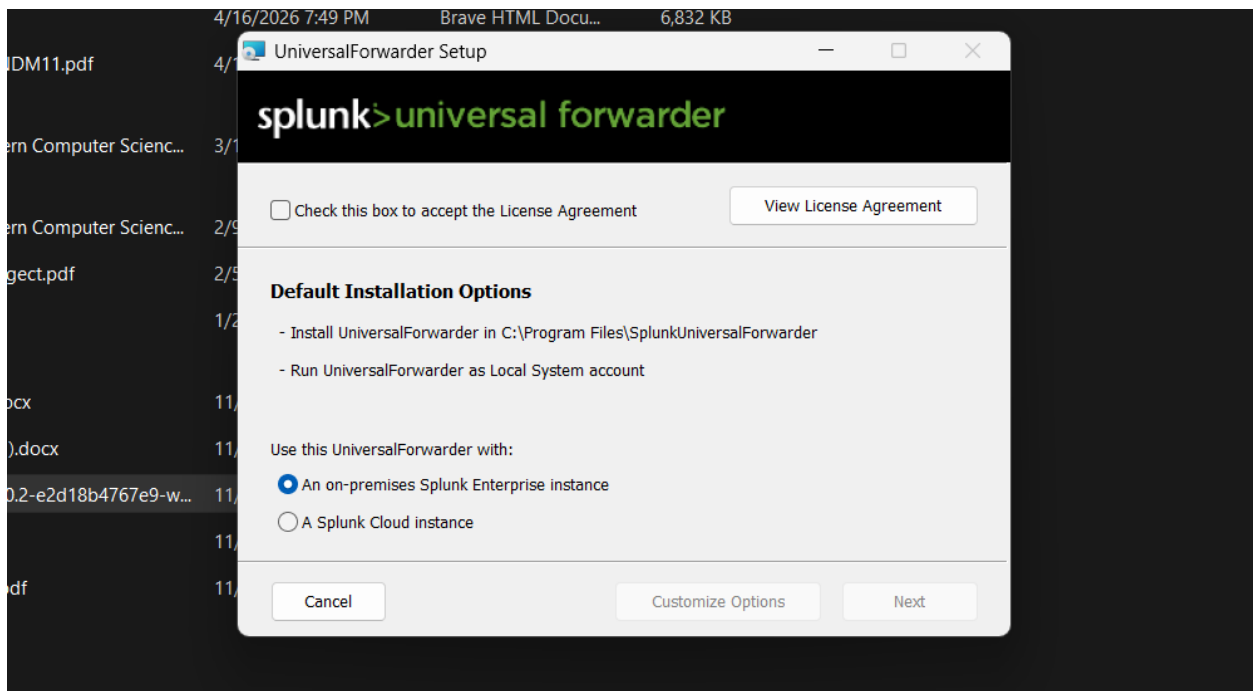
4.4.3 Installing Splunk Universal Forwarder (Windows)

This section describes the installation of the Splunk Universal Forwarder on a Windows machine, which is used to collect logs and forward them to the Splunk SIEM server.

Step 1: Accept License and Select Installation Type

Accept the license agreement and proceed with the default installation settings.

Select “On-premises Splunk Enterprise instance” as the deployment type.

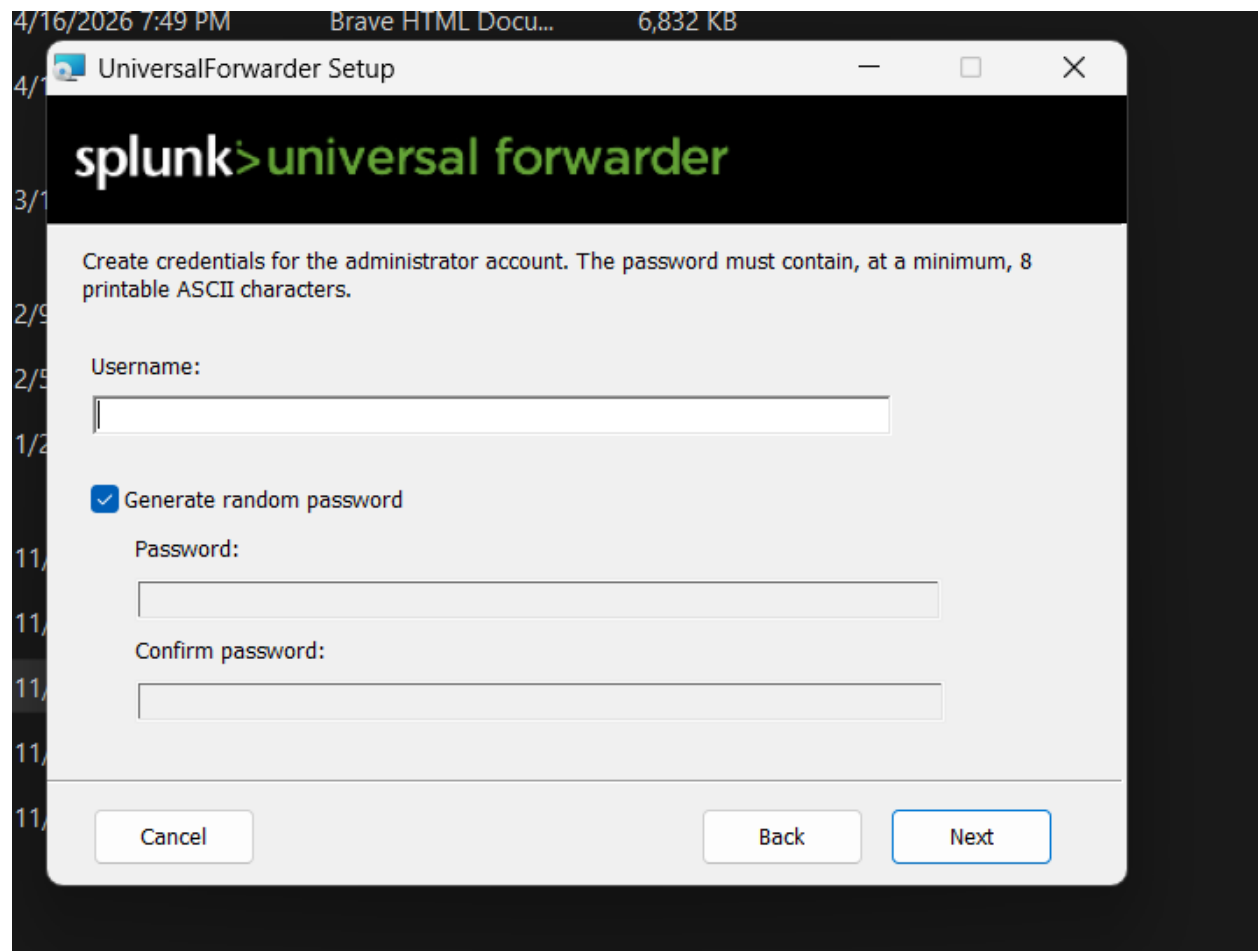


Step 2: Configure Administrator Credentials

Enter the administrator credentials as follows:

- Username: admin
- Password: Admin123

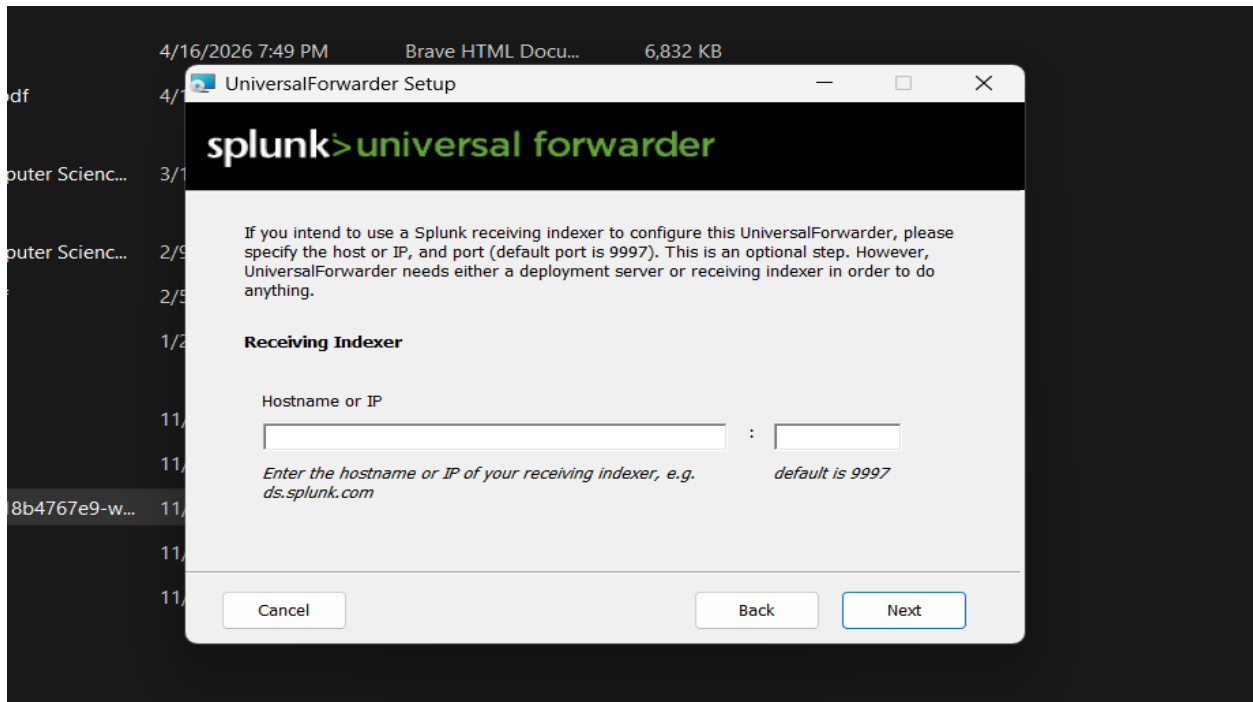
This account will be used to manage the forwarder locally.



Step 3: Configure Receiving Indexer

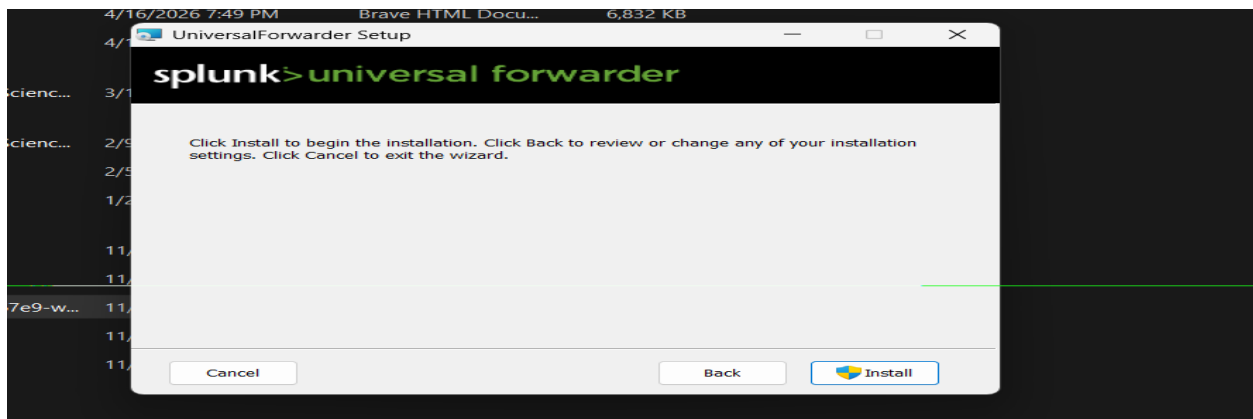
Enter the IP address of the Splunk server (Indexer) along with the default port 9997.

This step establishes the connection between the forwarder and the central SIEM system.



Step 4: Start Installation

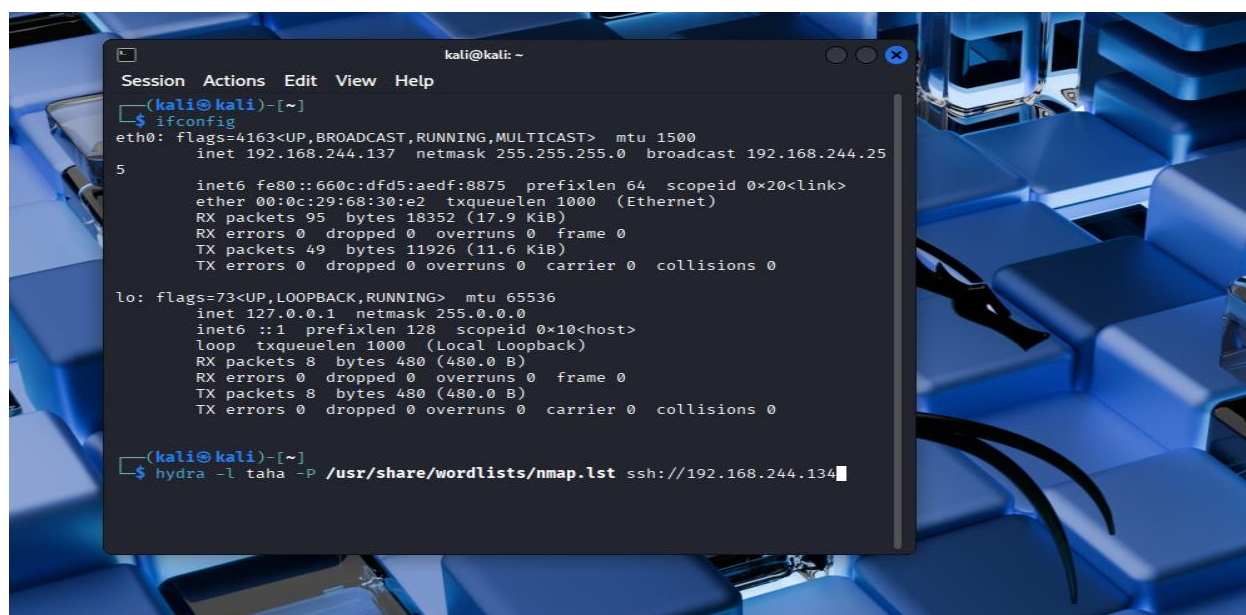
Review the configuration settings, then click Install to begin the installation process.



Successful Brute Force Attack Using Hydra

This figure illustrates the execution of the brute force attack using Hydra against the target system. The tool attempts multiple login combinations and successfully identifies valid credentials.

The output shows that the username “**taha**” and password “**Admin123**” were successfully discovered. This confirms that the attack was able to compromise the target system by exploiting weak authentication credentials.

A terminal window titled 'kali@kali: ~' is shown over a background of blue computer keyboard keys. The terminal displays the output of the 'ifconfig' command, showing details for the 'eth0' and 'lo' interfaces. Below this, the 'hydra' command is entered: 'hydra -l taha -P /usr/share/wordlists/nmap.lst ssh://192.168.244.134'. The command is partially executed, with a cursor at the end.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ifconfig  
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500  
    inet 192.168.244.137 netmask 255.255.255.0 broadcast 192.168.244.255  
    ether 00:c:29:68:30:e2 txqueuelen 1000 (Ethernet)  
    RX packets 95 bytes 18352 (17.9 KiB)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 49 bytes 11926 (11.6 KiB)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536  
    inet 127.0.0.1 netmask 255.0.0.0  
    inet6 ::1 prefixlen 128 scopeid 0<host>  
    loop txqueuelen 1000 (Local Loopback)  
    RX packets 8 bytes 480 (480.0 B)  
    RX errors 0 dropped 0 overruns 0 frame 0  
    TX packets 8 bytes 480 (480.0 B)  
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0  
  
(kali@kali)-[~]  
$ hydra -l taha -P /usr/share/wordlists/nmap.lst ssh://192.168.244.134
```

```
(kali㉿kali)-[~]
$ hydra -l taha -P /usr/share/wordlists/nmap.lst ssh://192.168.244.134 -t 2 -w 3
Hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organiza
tions, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

[WARNING] the waittime you set is low, this can result in erroneous results
Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-30 14:38:05
[DATA] max 2 tasks per 1 server, overall 2 tasks, 5010 login tries (l:1/p:5010), ~2505 tries per task
[DATA] attacking ssh://192.168.244.134:22/
[22][ssh] host: 192.168.244.134 login: taha password: Admin123
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-30 14:38:07

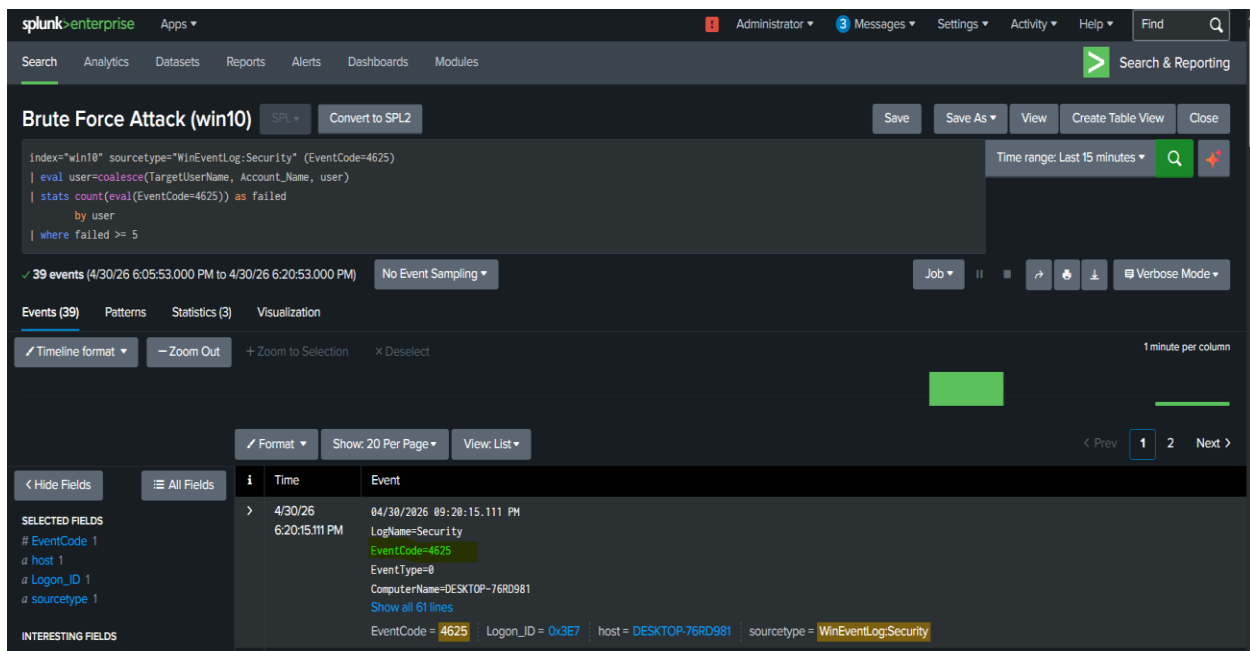
(kali㉿kali)-[~]
$
```

Detection of Brute Force Activity in Splunk

This figure shows the detection of the brute force attack within the SIEM platform, Splunk Enterprise.

A search query was executed on Windows Security logs using Event ID 4625, which represents failed login attempts. The query aggregates the number of failed logins per user and identifies abnormal behavior based on a predefined threshold.

The results demonstrate multiple failed authentication attempts, indicating a brute force attack pattern. This confirms that the SIEM system successfully detected the attack activity through log analysis.



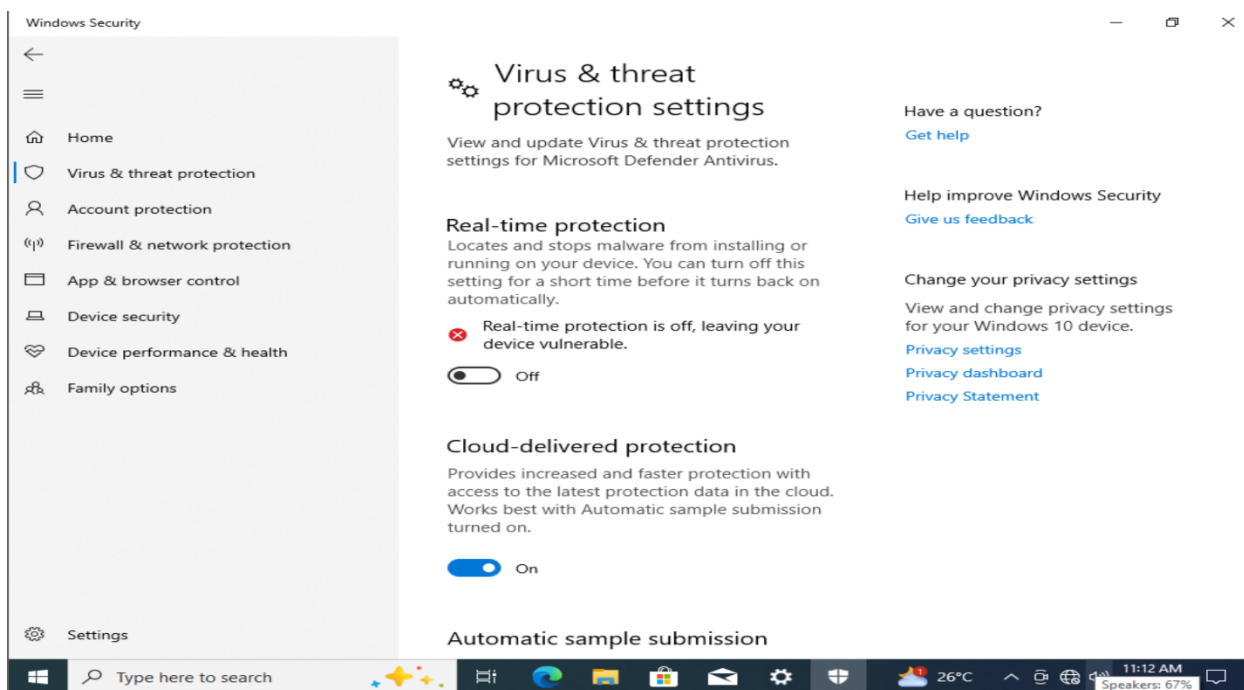
Detection of Defense Evasion Activity in Splunk

This figure shows the detection of defense evasion activity within the SIEM platform, Splunk Enterprise.

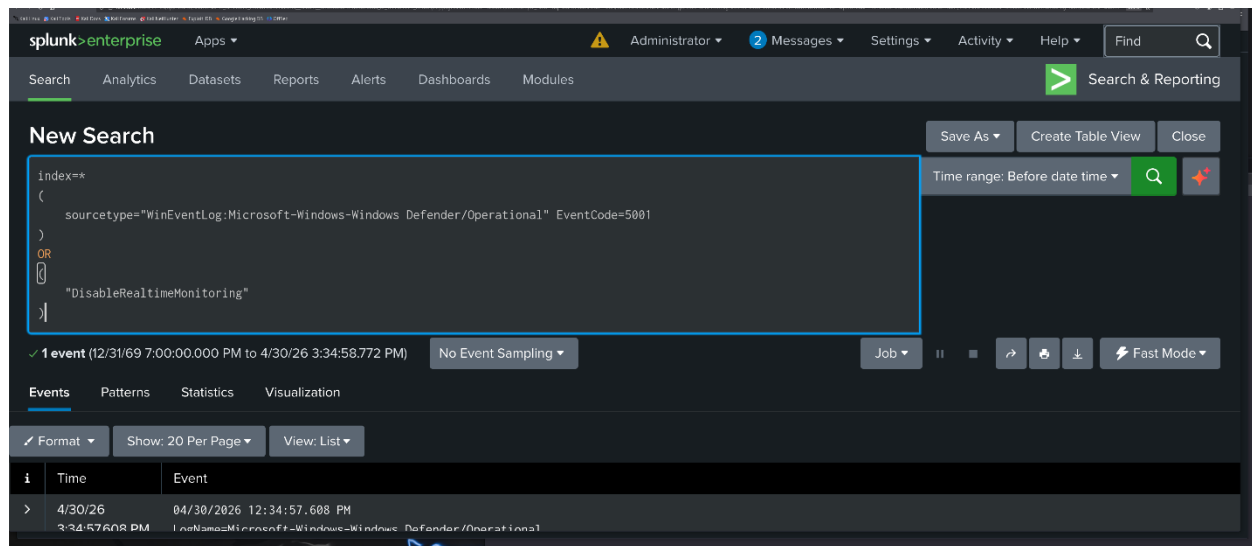
A search query was executed on Windows Defender operational logs using **Event ID 5001**, which indicates that Microsoft Defender real-time protection was disabled. The query also searches for the keyword **DisableRealtimeMonitoring**, which is commonly associated with attempts to turn off real-time monitoring through PowerShell or system configuration changes.

The results demonstrate suspicious activity related to disabling Windows Defender security features. This behavior is considered a defense evasion technique because attackers often attempt to disable antivirus or endpoint protection to avoid detection.

This confirms that the SIEM system successfully detected the defense evasion activity through Windows Defender log analysis and keyword-based monitoring.

A screenshot of the Splunk Enterprise 'Triggers Alerts' page. The page shows a list of triggered alerts with columns for Time, Alert name, App, Type, Severity, Mode, and Actions. The alerts are filtered by 'All alerts' and show 1-5 of 5 results. The alerts are as follows:

Time	Alert name	App	Type	Severity	Mode	Actions
2026-04-30 15:34:58 EDT	Defense Evasion - Defender Disabled	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
2026-04-30 15:34:44 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
2026-04-30 15:34:39 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
2026-04-30 15:33:39 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
2026-04-30 15:33:38 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete



Detection of Suspicious PowerShell Execution in Splunk

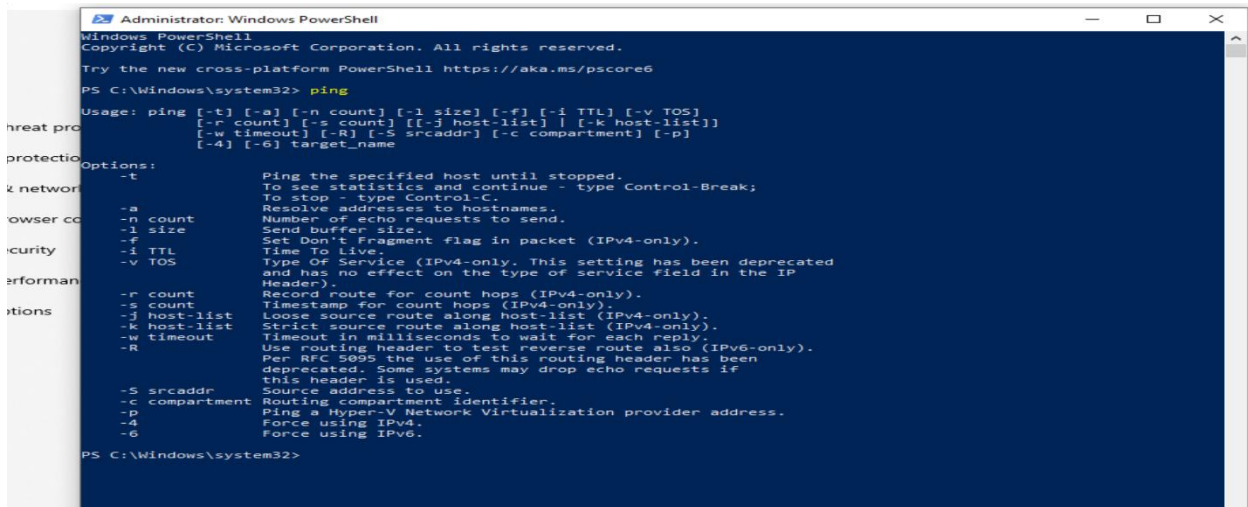
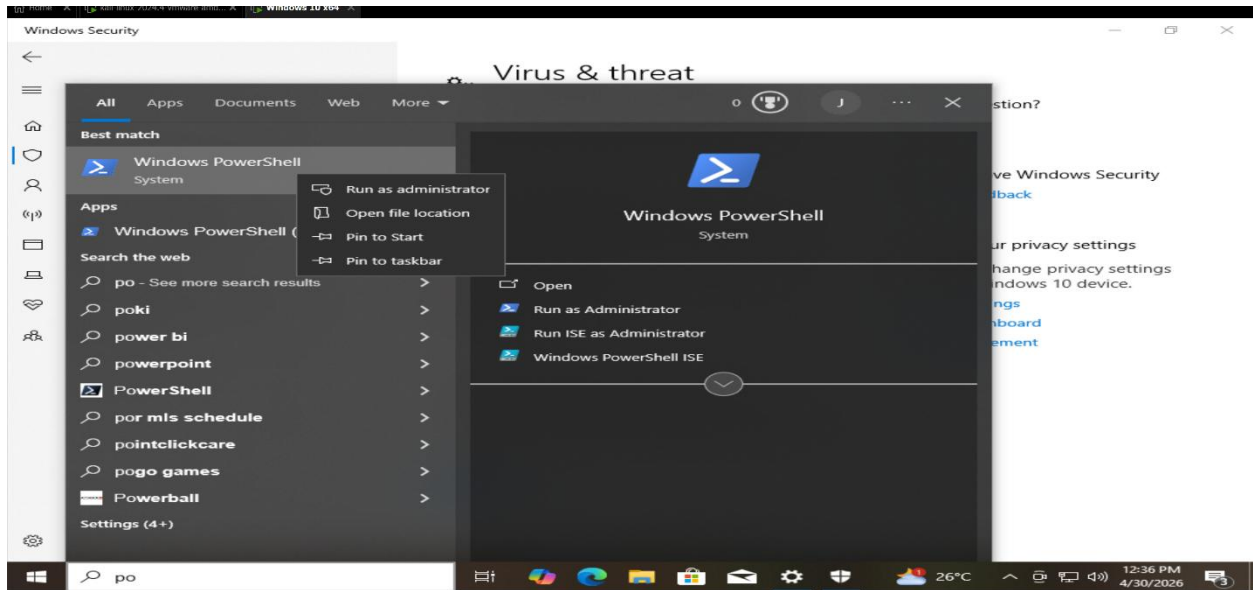
This figure shows the detection of suspicious PowerShell execution within the SIEM platform, Splunk Enterprise.

A search query was executed on Windows Security logs using **Event ID 4688**, which represents process creation events. The query filters the results to identify newly created processes where the process name contains **powershell.exe**.

The results demonstrate the execution of PowerShell on the monitored Windows system. PowerShell is a legitimate administrative tool, but attackers

commonly abuse it to execute commands, download payloads, run scripts, or perform post-exploitation activities.

This confirms that the SIEM system successfully detected suspicious PowerShell activity through process creation log analysis.



splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Triggered Alerts

Filter

App Search & Report... Owner All owners Severity All severity

Showing 1 - 6 of 6 results Show: 50 p... 1 of 1 pages

Alert name All alerts

☐	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2026-04-30 15:36:56 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
☐	2026-04-30 15:36:39 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
☐	2026-04-30 15:36:38 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
☐	2026-04-30 15:35:39 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
☐	2026-04-30 15:35:39 EDT	PowerShell Execution Detected	search	Real-time	Medium	Per Result	View Results Edit Alert Delete
☐	2026-04-30 15:34:58 EDT	Defense Evasion - Defender Disabled	search	Real-time	Medium	Per Result	View Results Edit Alert Delete

splunk>enterprise Apps Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards Modules Search & Reporting

New Search

Save As Create Table View Close

index=* sourcetype="WinEventLog:Security" EventCode=4688
| search New_Process_Name="powershell.exe"

Time range: Before date time

1 event (12/31/69 7:00:00.000 PM to 4/30/26 3:36:56.153 PM) No Event Sampling

Job

Events Patterns Statistics Visualization

Format Show: 20 Per Page View: List

i	Time	Event
>	4/30/26 3:36:54.822 PM	04/30/2026 12:36:54.822 PM LogName=Security EventCode=4688 EventType=0 ComputerName=DESKTOP-K7MKNKBD Show all 34 lines host = DESKTOP-K7MKNKBD source = WinEventLog:Security sourcetype = WinEventLog:Security